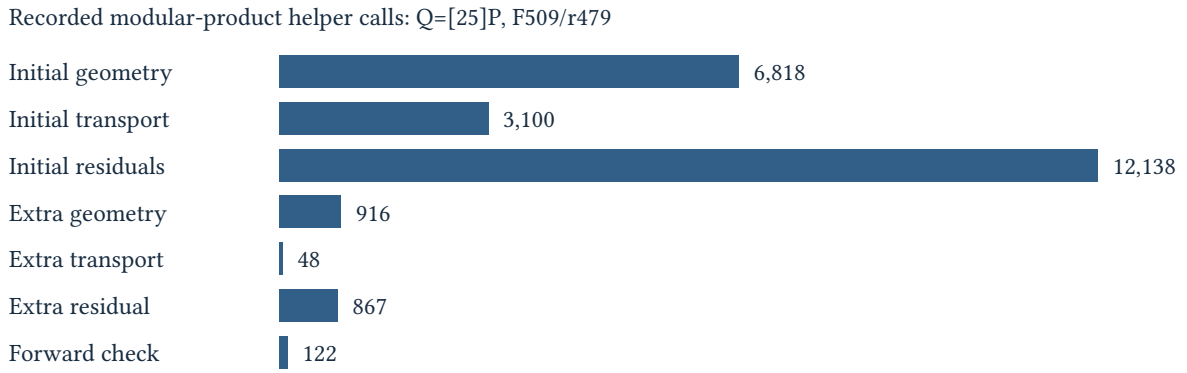
Normalized recovery: arithmetic and allocation

Executed 2026-11-05; reconciled 2026-11-07 · T-1727

The accounting instrument separates compiler/setup, geometry, normalized transport, character residuals, decoding and public verification. Fifteen source modules have reversible instrumentation maps; counters cover modular-product helpers, inverses, popcounts, rule tests and allocator events. [pcc:unverified]



Recorded breakdown

For this saved fixture, fourteen initial residuals record 12,138 product-helper calls: 7,714 polynomial and 4,424 source operations. Initial transport records 3,100: 2,250 phase, 488 value and 362 transport. Two extra geometry steps cost 916 helpers; extra transport 48; the one extra residual 867; public verification 122. These are helper-event counts, not time weights. [pcc:unverified]

Compilation records 24,076,995 products, calibration 5,974 and preparation 818, separately from queries. Initial-character allocation for the displayed query requests 218,526 cumulative bytes. Requested and freed bytes are event volumes, not peak or net memory. The original compiler-version string and host inventory were not recorded. [pcc:unverified]

Instrument validation and custody

Three accounting tests pass; 414 inherited tests are filtered. All 20 public-point outputs agree with frozen normalized recovery and pass forward multiplication. Scope restoration, thread isolation, allocator forwarding, phase/value initialization placement and skipped-character accounting are checked. [pcc:unverified]

Fifteen forward/reverse transforms pass and 30 byte corruptions are rejected. The 53 source/data hashes still match. Projection contains all 1,140 unique raw count records, including the first test-prefixed row. The native receipt and logs record completion on November 5. The old process handle was absent on November 7; native execution was not repeated. Sixty compiler warnings are retained. [pcc:unverified]

Next construction and scope

The residual/polynomial breakdown and allocation volume motivate combined-monomial transport and geometry-only residual simplification. A matched whole-path comparison against batch reciprocals remains to be bound. This fixture has one counted candidate and an uninstrumented correctness reference. Adds, branches, moves, hashing, some inline arithmetic and machine costs are outside the counters. No timing or cloud panel; original 4,563-case cohort unchanged.

Claim PCC-20261107-0f3d29 remains unsigned (local verdict not_covered). Sources checked: saved batch/quartic instrumentation, meter, projection parser and normalized recovery source. KB unavailable (MCP absent, wrapper executable missing). Evidence: raw logs, accounting-receipt.json, instrumentation-check.json, projection-check.json and counts.json.